

Simple Machines

16

Look here guys! I made a mini vehicle. Isn't it great?

Indeed I did!

That looks pretty amazing. Did you incorporate the concepts of stability and balance for this?



Machines help us to clean, move objects, create and repair things, relax, and have fun. Simple machines are the most basic machines that we use to make our work easier in our daily lives. All the complicated machines we use today are based on simple machines from earlier times.

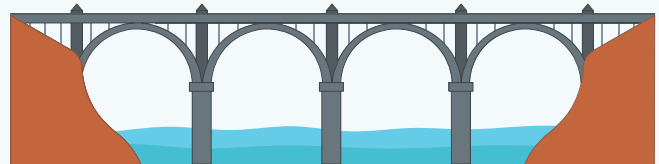
Exercise

Answer the following questions:

- 1 Do you think a human being can lift the same amount of weight as a machine can?



- 2 Do you know why bridges are so strong?



Structures:

A structure is the alignment or organisation of different elements to create a model. A structure is built by putting many parts together. It can be anything, like a building, bridge, or pyramid.

Stability:

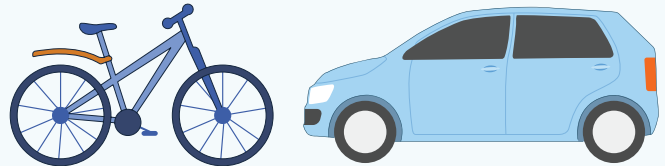
The stability of a structure is its ability to maintain its form, and not move, fall, or break. A stable structure stays unaffected under different weather conditions, Pressure, weight, height, wind velocity etc.

For example: High-rise buildings, long bridges, etc.

Exercise

Answer the following questions:

- 3 Which of the following is more likely to fall?



Balance

Balance refers to an even distribution of weight that enables someone or something to remain upright and steady. If the weight of a structure is not evenly distributed, then it is an imbalanced structure. An imbalanced structure has higher chances of falling. Any pressure applied on the structure must be equally distributed.

For example, a table.



A table has 5 parts: 4 legs and 1 surface. All of these parts help it remain stable and balanced. If we remove any one part, the table will become unstable and fall.

Similarly, for a building to be strong and tall it needs to be stable and balanced. Have you ever wondered how humans managed to build so many big and heavy things with minimum human efforts? It was made possible because of the simple machines we used.

Simple machines

A simple machine is a mechanical device with few or no moving parts which makes our work easier.

There are six types of simple machines:

a. Pulley:

A pulley is a combination of one or more wheels, over which we loop a rope to make it easier to lift things.



Pulley

b. Wheel and Axle:

It is a type of simple machine that uses one or more wheels with a rod attached in the middle. The rod acts as an axle to help it to lift or move loads.



Wheel and axle

c. Inclined Plane:

An inclined plane is simply a flat surface raised at an angle. This is used to move objects from a lower point to a higher point, or vice versa, with less effort.



Inclined plane

d. Screw:

A screw is an inclined plane wrapped around a pole. It is used to lift things or hold them together.



Screw

e. Wedge:

Two inclined planes placed back to back make a wedge. A sideways force is generated when a wedge is pushed between two objects. It is usually made of metal or wood, and is used for splitting, lifting, or tightening objects.



Wedge

f. Lever:

A lever helps to lift objects with less effort. It is a long tool, such as a pole or a rod, put under an object with a support to lift it.



Lever

In this lesson, we will build various models of levers that use two different types of simple machines: inclined planes and wheel and axles.

Let us understand levers in detail.

Lever: A lever is a bar that rests on a support, called a fulcrum. A downward force exerted on one end of the lever allows a small force to lift a heavy weight at the other end.

Bytes

A famous quote by Archimedes says – “Give me a lever long enough and a fulcrum on which to place it, and I shall move the world”.

Parts of a lever

There are three main parts of a lever: **fulcrum**, **load**, and **effort**.

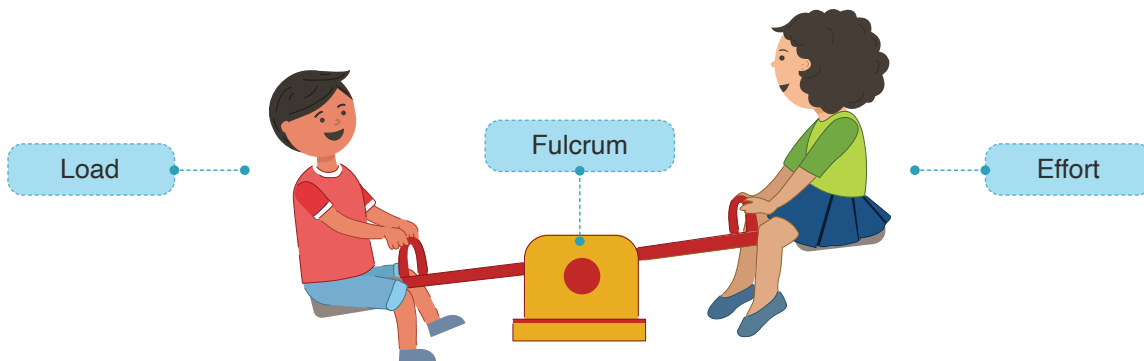
The object to be moved is the load. The force applied to move the load is the effort. The fulcrum (or pivot) is the point used to support the lever.

For example, a man can lift a stone with a lever.

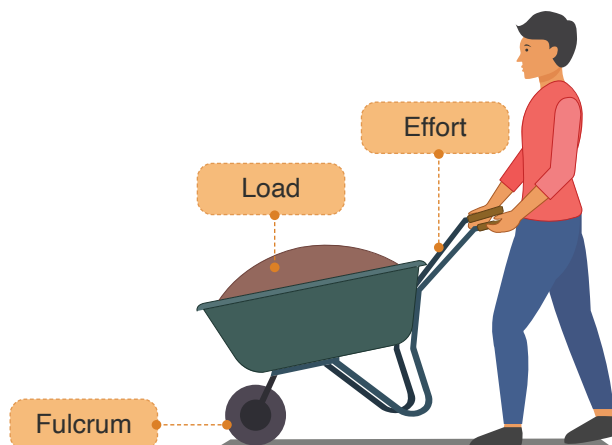


Based on the location of the **fulcrum**, **load**, and **effort**, levers are divided into three types.

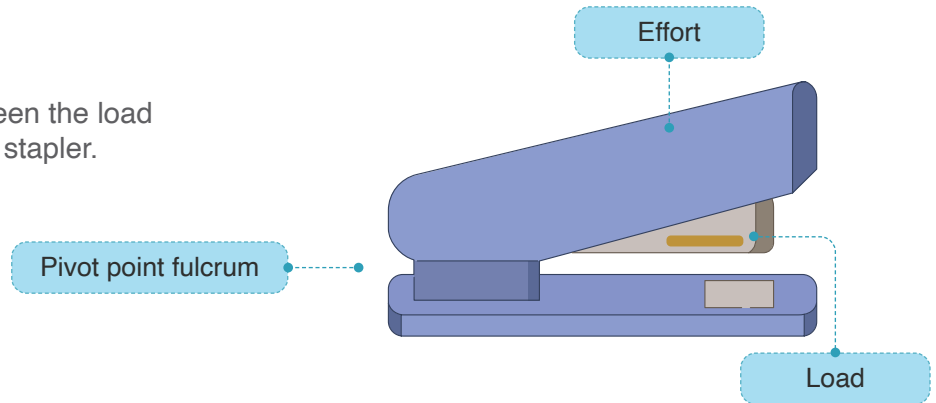
a. Class 1 lever: The fulcrum is between the effort and the load. For example, a see-saw.



b. Class 2 lever: The load is between the effort and the fulcrum. For example, a wheel barrow.

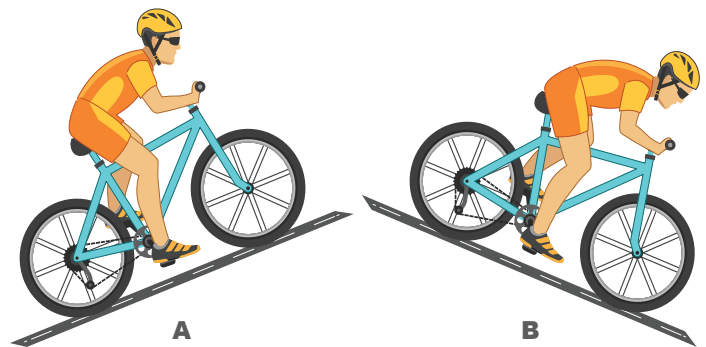


c. Class 3 lever: The effort is between the load and the fulcrum. For example, a stapler.



There is one more machine that we use daily, through which we can achieve our goal much faster and with less effort.

Observe the following pictures carefully and decide which bicycle requires more power and which has more speed? Why?



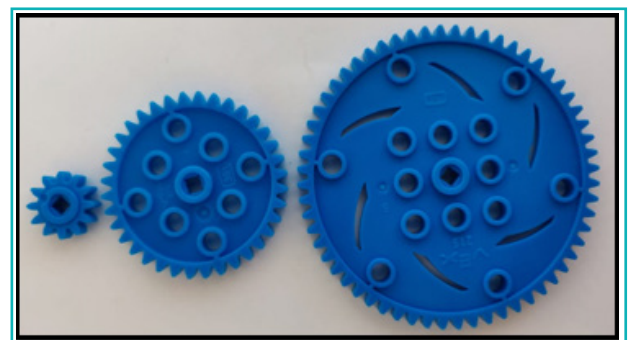
The first bicycle will need more power because it is going uphill. The other bicycle will have a higher speed because it is going downhill. Both these requirement can be fulfilled using gears.

Gears

Gears are mechanical parts with cut teeth. They fit into another toothed part to transmit or receive force or action.

Gears are generally used for one of three different reasons:

1. To reverse the direction of rotation
2. To increase or decrease the speed of rotation
3. To transfer rotational motion to a different axis



The gear structure helps to move fans at higher speeds and less efforts, with respect to its gear ratio.

Gear Ratio

Gear Ratio is the relationship between a Driving Gear (the gear connected to the input power source, such as a motor or lever) and a Driven Gear (the gear connected to the output, such as a wheel or fan blades) in a system.

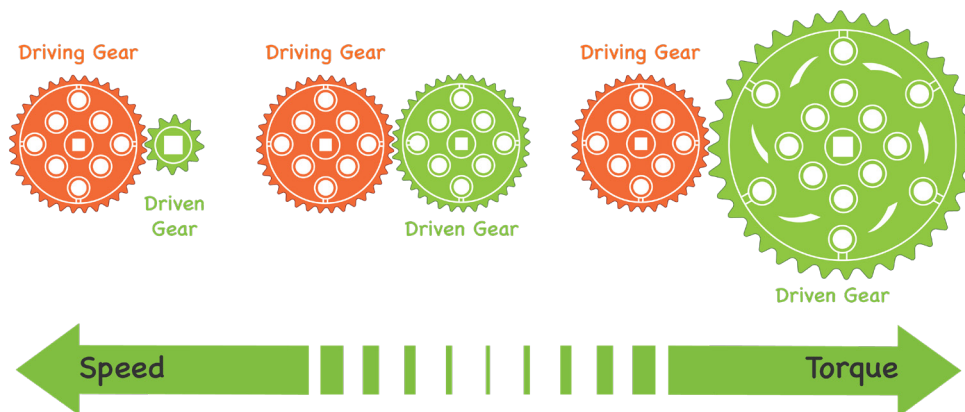
Gear Reduction

When a system has a driving gear that is SMALLER than the driven gear, it will increase the Torque and decrease Speed.

Speed is the total distance covered, divided by the total time taken.

Speed = total distance covered / total time taken

Torque is the measure of the force that causes an object to rotate about an axis.



Making this kind of change to a mechanical structure is helpful when we try to lift or move faster mechanically, we don't require the ability to lift heavy objects.

Gear Ratio is expressed this way:

(Driving Gear Teeth): (Driven Gear Teeth)

Gear Reduction is expressed in reverse:

(Driven Gear Teeth): (Driving Gear Teeth)

Which means for example, if driving gear has 12 teeth and driven gear has 60 teeth, then the Gear Ratio will be:

$$12:60 = 1:5$$

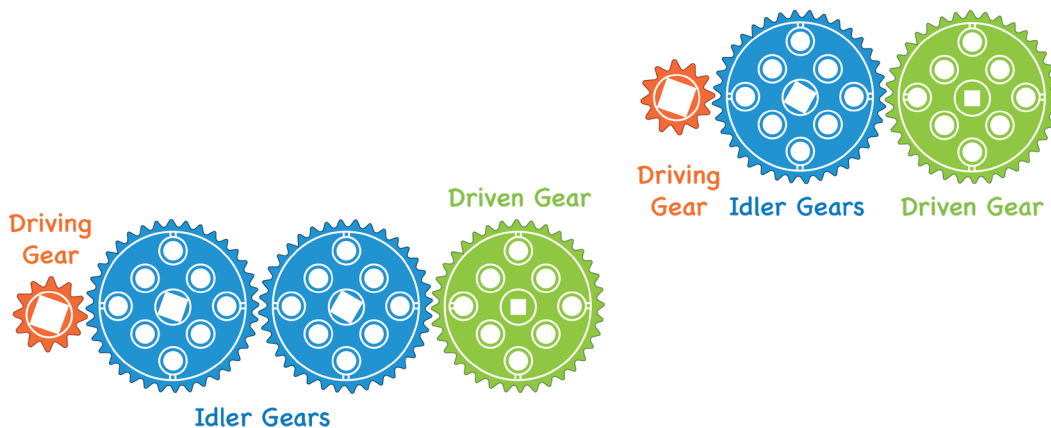
Whereas, gear reduction will be:

$$60:12 = 5:1$$

i.e. for every 1 rotation of driving gear with 60 teeth, there will be 5 rotations of driven gear with 12 teeth.

Idler Gears

- A simple Gear Train is a connected set of rotating gears that transmit power from an input (like a driving gear connected to a motor) to an output (like a driven gear connected to a wheel or mechanism).
- Simple gear trains can have any number of gears in a single row.
- All gears in between the driving gear and the driven gear that only transmit power are known as Idler Gears.
- Idler gears have NO EFFECT on gear ratio or gear reduction, regardless of the number of teeth they have.
- Idler gears can be used to change the power direction from driving gear to driven gear.

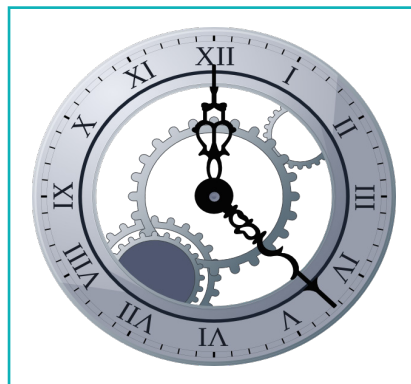


Applications of Gears

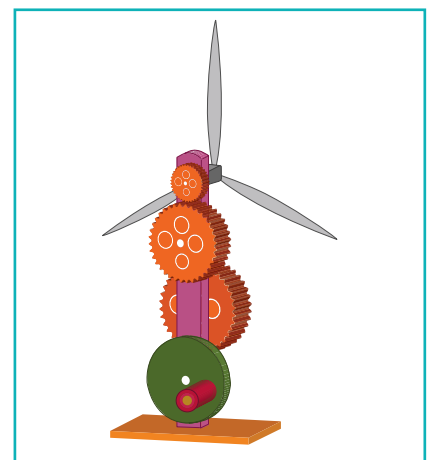
a. Bicycle



b. Analog (non-digital) clocks



c. Windmill



Exercise

Fill in the blanks

- 4 A lever is a bar that rests on a support called a _____ .
- 5 The _____ of bridges provide them with stability, which make them strong.
- 6 _____ refers to an even distribution of weight enabling someone or something to remain upright and steady.
- 7 _____ is the relationship between a Driving Gear and a Driven Gear in a system.

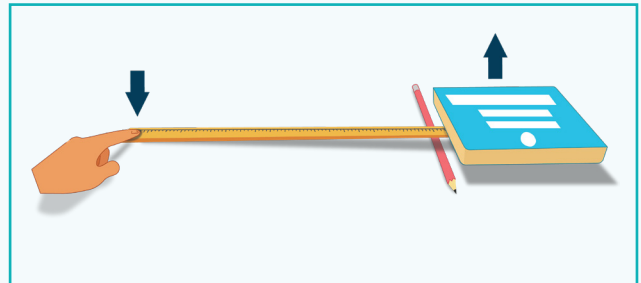
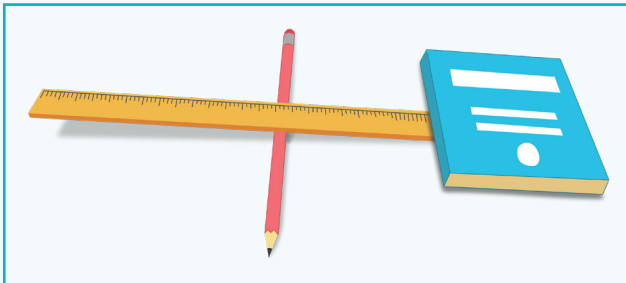
State whether True or False

- 8 The stability of a structure is its ability to maintain its form, and not move, fall or break.
▲ _____
- 9 An inclined plane is a combination of one or more wheels with a looped rope, to make it easier to lift things.
▲ _____
- 10 Torque is a measure of the force that can cause an object to rotate about an axis.
▲ _____
- 11 Idler gears affect the Gear Ratio or Gear Reduction.
▲ _____

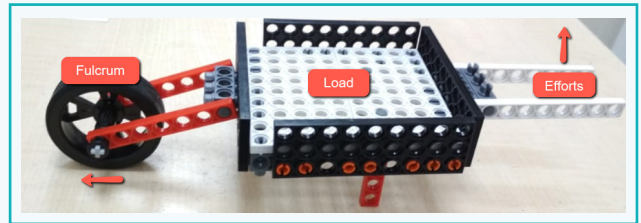
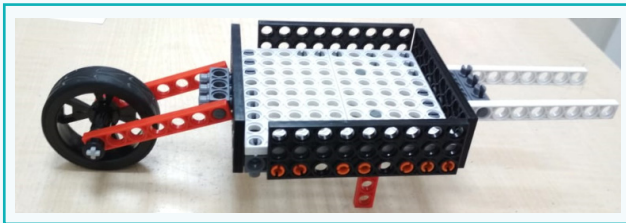
Activity

- Try to construct the models of various levers using readily available things around you (for example, ruler, pencil, eraser, book, etc.). You can also use pre-moulded kits like (VEX IQ kit)

a. Class 1 lever



b. Class 2 Lever



Today, we learnt about simple machines. Since the beginning of time, humans have developed devices and tools to make our work easier. The most notable among these are the six types of simple machines: the inclined plane, the wheel and axle, the pulley, the lever, the screw, and the wedge.

Engineers creatively use their knowledge to make our lives better, often using simple machines. They invent tools that make work easier. They accomplish huge tasks that would not be done without the mechanical advantage of simple machines. Engineers design structures and tools to use our environmental resources better and more efficiently.



Tech Flash

- **Structure** : A structure is the alignment or organization of different elements to create a model.
- **Stability** : The stability of a structure is its ability to maintain its form, and not move, fall or break.
- **Balance** : Balance refers to an even distribution of weight enabling someone or something to remain upright and steady.
- **Simple Machine** : A simple machine is a mechanical device that makes our work easier using a few things to complete a task faster. There are six types of simple machines: Pulley, Lever, Wheel and Axle, Screw, Wedge, and Inclined Plane.